

1. Introduction

The Seller and Inventory Management System is an extension of the Product and Category Management System developed in Task 2. The system manages seller details, product assignments and inventory stock information. It uses relational database concepts including primary keys, foreign keys, NOT NULL, UNIQUE and CHECK constraints. SQL queries are used to perform CRUD operations and generate inventory analysis reports.

2. Seller Table Design

Field	Data Type	Constraint
Seller_ID	INT	Primary Key
Seller_Name	VARCHAR(50)	NOT NULL
Email	VARCHAR(100)	NOT NULL, UNIQUE
Phone	VARCHAR(15)	NOT NULL, UNIQUE
Address	VARCHAR(200)	NOT NULL

3. Inventory Table Design

Field	Data Type	Constraint
Inventory_ID	INT	Primary Key
Product_ID	INT	Foreign Key, NOT NULL
Seller_ID	INT	Foreign Key, NOT NULL
Stock_Quantity	INT	NOT NULL, CHECK
Stock_Status	VARCHAR(20)	NOT NULL, CHECK
Last_Updated	DATE	NOT NULL

4. Updated ER Diagram

Put your ER diagram here.

The relationship is:

Category



Product



Inventory



Seller

More accurately:

Seller 1 ————— M Inventory M ————— 1 Product

5. Sample Seller Data

Use:

```
SELECT * FROM Seller;
```

Take a screenshot and put it here.

Expected data includes:

Seller_ID	Seller_Name	Email	Phone	Address
1	Tech World	techworld@gmail.com	9876543210	Chennai
2	Smart Mart	smartmart@gmail.com	9876543211	Coimbatore
3	Digital Hub	digitalhub@gmail.com	9876543212	Madurai
4	Gadget Store	gadgetstore@gmail.com	9876543213	Thoothukudi
5	Online Bazaar	onlinebazaar@gmail.com	9876543214	Tirunelveli

6. Sample Inventory Data

Use:

```
SELECT * FROM Inventory;
```

Take a screenshot.

Your report can show:

Inventory_ID	Product_ID	Seller_ID	Stock	Status
1	1	1	25	Available
2	2	1	8	Available
3	3	2	0	Out of Stock
4	4	2	15	Available
5	5	3	5	Available

7. Seller Operations

Include screenshots of:

Add Seller

```
INSERT INTO Seller ...
```

Assign Product

```
INSERT INTO Inventory ...
```

Products supplied by each seller

```
SELECT  
  s.Seller_Name,  
  p.Product_Name,  
  i.Stock_Quantity
```

```
FROM Seller s
JOIN Inventory i
ON s.Seller_ID = i.Seller_ID
JOIN Product p
ON i.Product_ID = p.Product_ID;
```

Count products

```
SELECT
    s.Seller_Name,
    COUNT(i.Product_ID) AS Product_Count
FROM Seller s
LEFT JOIN Inventory i
ON s.Seller_ID = i.Seller_ID
GROUP BY s.Seller_ID, s.Seller_Name;
```

Update seller

Show the UPDATE query and its output.

8. Inventory Operations

Include screenshots for:

Available Products

```
WHERE i.Stock_Quantity > 0
```

Out of Stock

```
WHERE i.Stock_Quantity = 0
```

Stock Less Than 10

```
WHERE i.Stock_Quantity < 10
```

Update Stock

```
UPDATE Inventory
SET Stock_Quantity = 20
WHERE Inventory_ID = 2;
```

Delete Discontinued Inventory

```
DELETE FROM Inventory
WHERE Inventory_ID = 6;
```

9. Inventory Analysis Report

Include these six outputs:

1. Seller-wise Product Report

Shows which seller supplies which product.

2. Stock Availability Report

Shows the quantity and status of every inventory item.

3. Total Available Products

```
SELECT COUNT(*) AS Total_Available_Products
FROM Inventory
WHERE Stock_Quantity > 0;
```

4. Out-of-Stock Products

Shows products where stock is 0.

5. Highest Stocked Product

```
ORDER BY Stock_Quantity DESC
LIMIT 1;
```

6. Average Inventory Quantity

```
SELECT AVG(Stock_Quantity)
FROM Inventory;
```

10. Conclusion

You can directly put this in your PDF:

Conclusion:

The Seller and Inventory Management System was successfully implemented

using MySQL. The system manages seller information, product assignments and inventory details. Primary key, foreign key, NOT NULL, UNIQUE and CHECK constraints were implemented to maintain data integrity. CRUD operations were performed on seller and inventory data. SQL joins and aggregate functions were used to generate seller-wise and inventory analysis reports such as available stock, out-of-stock products, highest stocked product and average inventory quantity. The project demonstrates the practical application of relational database management concepts.